

Tank Survival Game Design Document

By Sheldon Hung



Game Overview

Tank Survival Game is about surviving waves of various enemies. The player survives by operating a tank and using its weapons to defend themselves. To win the game, the player must survive three waves of enemies.

Game Link: <https://sheldon-hung.itch.io/tank-survival>

Github: <https://github.com/sheldonwuhung/Tank-Survival-Labs>

Target Audience

Demographics

This game is directed towards people who are fans of shooters or warfare games because it involves military vehicle combat.

Difficulty

People who are comfortable with top-down shooters might find this game to be easy, and those unfamiliar will find it more difficult.

Concept

Game Idea

The concept of the game is wave survival and each succeeding level gets progressively more difficult. The number of enemies will be greater and more difficult enemies will also be introduced.

Goals

The goal of the game is to survive all three waves of enemies. Players can earn points by defeating enemies, and if they fail and restart a wave, they lose points. Players may have the goal of earning as many points as possible while avoiding failure.

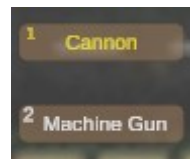
Controls & Mechanics

Movement

The player controls the movement of the tank by using WASD or arrow keys. Aiming a weapon is controlled by the mouse or trackpad, and firing the weapon is done with a left mouse click. Press Escape to pause.

Weapon Switching

Players can choose from two weapons by pressing 1 or 2. The cannon does more damage and has a long cooldown and the machine gun does less damage and has a quick cooldown.



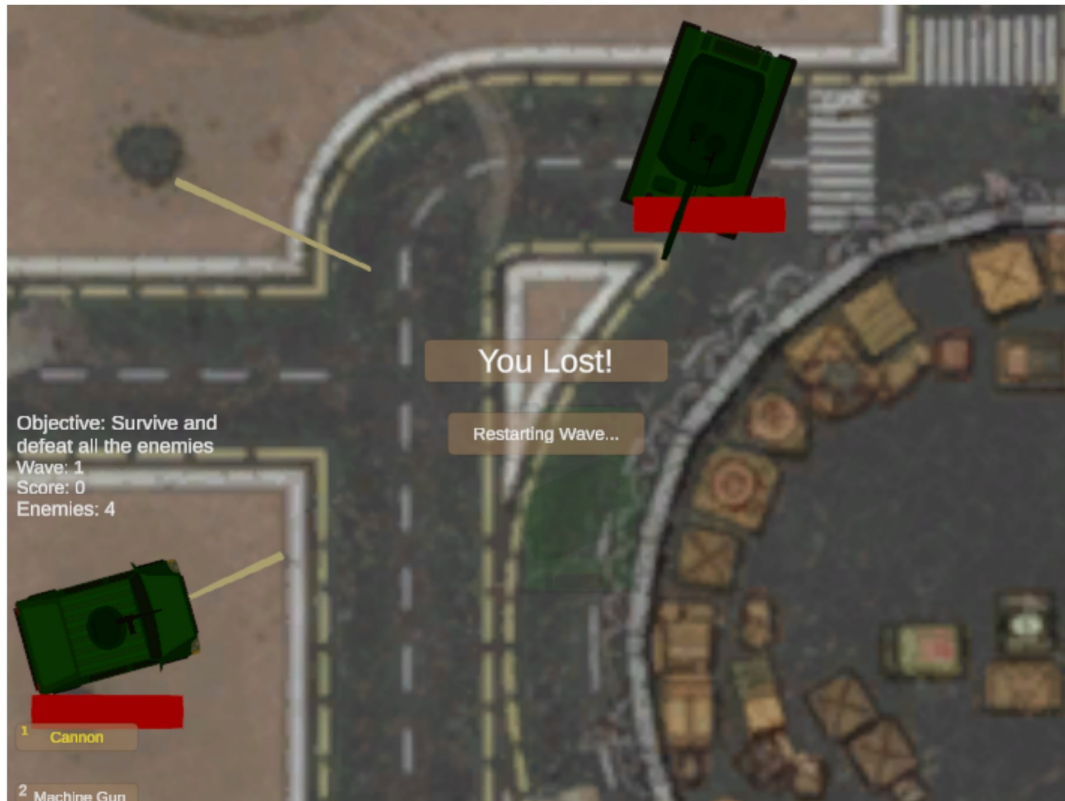
Health

Both players and enemies have health indicated by a red health bar under their sprite. For players to fail, they must lose all their health. For enemies to die, they must lose all their health. When enemies die, they drop a repair box that when touched by the player, restores health. Health is carried over between each level to provide challenge and to get players to think about managing resources.



Level Progression

Players must defeat all enemies in the current wave in order to progress to the next wave. If players lose all health from other enemies shooting at them, then they fail and have to repeat the level, losing points. If they defeat enemies, they earn points and if they defeat all enemies within the wave, they progress to the next one.



Enemies

Behavior

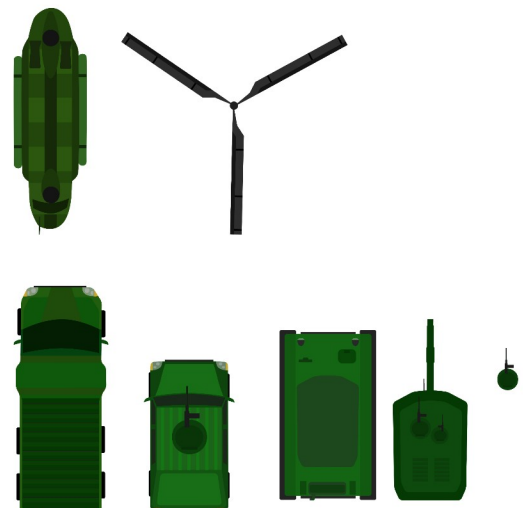
Enemies will start shooting when they are within a certain distance from the player. They will stop following the player at a certain distance, and if the player gets too close, they will attempt to move backwards.

NPC Types

Jeep: Shoots bullets at the player and does very little damage at a high fire rate.

Tank: Shoots cannon rounds at the player and does more damage at a slow fire rate.

Chopper: Shoots missiles at the player and does high damage at a slow fire rate.



Scoring

High Scores

High scores are recorded whenever a player beats all three levels. Players can attempt to obtain the highest score by not dying in any of the levels as dying means losing points.

Known Issues

- Fire rate delay when switching from cannon to machine gun.
- Player fires once when game is paused.
- If enemies are quickly killed, it can break the game loop.
- Chopper health bar stays after chopper gameobject is destroyed.
- Enemy tanks sometimes stop shooting sometimes.
- High scores record in reverse order (5 being first and 1 being last).
- Bullets make a weird rotating movement sometimes

Potential Implementations

- Group boiding system that ensures npcs do not bunch up. Will make their movement more strategic and realistic.
- Different weapons and player upgrades.
- Making the machine gun automatic.

Art & Sound Sources

Sprites: <https://turbo-developement-team.itch.io/2d-top-down-vehicles>

Sound Effects: <https://pixabay.com/sound-effects/search/tank/>

Map: AI generated by Gemini

Scripting Help & Documentation

Saving & Loading Data: <https://docs.unity3d.com/ScriptReference/PlayerPrefs.html>

Tweening: <https://dotween.demigiant.com/>

Getting Angle Between Two Vectors (Gemini): `turret rotation unity math atan2`

Button Interaction:

<https://docs.unity3d.com/530/Documentation/ScriptReference/UI.Button.onClick.html>

Health Bar Tutorial:

<https://discussions.unity.com/t/how-to-animate-a-basic-health-bar/906071>